

A Knowledge-Graph Framework for Transformer-Level Planning under Residential Technology-Adoption Scenarios

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Abstract

Residential electrification requires planners to connect building-stock evidence, technology-adoption assumptions, and distribution-network assets, but those information layers are normally fragmented. This paper presents the National Energy Digital Twin (NEDT), an ontology and knowledge-graph framework for representing that planning context. Its core contribution is an OWL 2 DL extension that connects archetype cohorts, technology states, scenario operators, demand-profile references, LV–MV–HV assets, capacity indicators, and provenance concepts. The extension reuses and maps to BOT, SAREF, SARAGON, GeoSPARQL, QUDT, SOSA/SSN and PROV-O, and is accompanied by SHACL shapes and competency-question patterns.

The Irish demonstration represents 2.14 million dwellings as station-level archetype cohorts. An allocate-then-maximise computational layer produces transformer-level residential-demand and capacity-screening indicators for 37,287 MV/LV station catchments. A re-generated Scenario 1 graph contains 764,115 triples; it passes SHACL validation and four executed competency questions retrieve scenario assumptions, 26 county inventories, 3,378 model-screened over-capacity stations, and KPI provenance. The case study demonstrates an auditable semantic integration and conditional infrastructure exposure-screening workflow, rather than a validated prediction of observed transformer overload.

Keywords: Distribution network planning, Knowledge graph, Ontology, Heat pump adoption, Infrastructure exposure screening, Residential electrification

1. Introduction

Residential energy decarbonisation depends mainly on two end-use transitions: replacing fossil-fuel boilers with heat pumps and electrifying transport. Both transitions shift demand from gas, oil, or petrol to the electricity network, and much of this additional load appears at the low-voltage end of the distribution system. Ireland's Climate Action Plan targets

large-scale deployment of heat pumps and electric vehicles by 2030 [1]. Across Europe, the recast Energy Performance of Buildings Directive, the Renewable Energy Directive, and national electrification strategies create similar pressures on residential distribution networks. The planning challenge is therefore not only growth in national electricity demand. Network planners must also identify where new residential demand will concentrate on MV/LV transformers, when those transformers will become stressed, and which technology transition drives the stress. If planners wait until overloads are observed, reinforcement can require three to seven years for civil and equipment works, and reactive intervention can cost substantially more per unit of capacity than planned reinforcement based on forward-looking demand intelligence [2].

Urban building energy modelling (UBEM) provides the established method for estimating building-stock energy demand at district, city, and national scales [3, 4]. Bottom-up UBEM workflows typically group buildings into archetypes using dwelling type, floor area, construction period, heating fuel, occupancy, weather zone, and energy-performance class, and then scale simulated demand profiles using census, survey, or building-energy records [5, 6]. These models support building-stock demand estimation, retrofit assessment, policy analysis, and long-term energy-system scenario studies [7, 8, 9]. However, UBEM workflows usually operate on administrative, statistical, or urban-planning spatial units rather than electricity-network catchments. This spatial mismatch limits their usefulness for distribution planning: a county, district, census tract, or small-area demand estimate does not directly show which MV/LV transformer must serve the additional demand or whether that asset has enough spare capacity [10]. Advanced-metering and monitoring approaches can detect emerging overload patterns from measured data [11], but they identify stress after load has already appeared and depend on monitored assets. This limits their value for proactive planning, especially in rural feeders where oil-heated dwellings may transition rapidly to heat pumps but where monitoring coverage can be weaker.

A second body of literature uses Linked Data, ontologies, and knowledge graphs to integrate fragmented building, energy, and infrastructure data. Semantic Web technologies provide a formal mechanism for representing heterogeneous entities, relationships, constraints, and provenance across datasets that were not originally designed to interoperate [12, 13, 14]. Early BIM ontology research showed that OWL-based representations can encode building objects, spaces, and design-stage constraints in machine-readable form [15, 16]. Niknam and Karshenas [17] proposed a shared ontology approach for semantic BIM data representation and demonstrated how graph structures can support information exchange across distributed construction knowledge bases. Pauwels and Terkaj [18] translated IFC into OWL through ifcOWL, establishing an important foundation for semantic BIM and linked building data. Beetz et al. [19] and Pauwels et al. [20] further demonstrated how IFC, RDF, and OWL can support building-information interoperability. These studies show that ontologies can

formalise building information, but their focus remains primarily on BIM data exchange, building-object representation, and construction information management rather than residential energy-system planning.

The Building Topology Ontology (BOT) provides a lighter and more modular alternative to full IFC-to-OWL translation. BOT represents sites, buildings, storeys, spaces, zones, building elements, and their topological relationships [21]. Schneider et al. [22] and Rasmussen et al. [23] showed how BOT can act as a central topology layer for linked building data, allowing other domain ontologies to attach product, sensor, energy, or operational information to building topology. This modular design is useful for digital-twin workflows because it avoids the full complexity of IFC while retaining essential spatial and topological relationships. However, BOT does not model residential archetype demand, heat-pump or EV adoption, transformer catchments, network capacity, or overload indicators. It provides an important reusable building layer, but it does not solve the distribution-network planning problem addressed in this paper.

Other ontologies focus on smart-building devices, systems, sensors, and operational metadata. Brick provides a unified metadata schema for building equipment, sensors, points, locations, and relationships, enabling portable smart-building applications and analytics [24, 25, 26]. SAREF provides a reference ontology for smart appliances and energy-related functions, while SAREF4ENER and SAREF4BLDG extend these ideas toward energy devices, demand-response functions, and building systems [27, 28, 29]. SOSA/SSN supports observation and sensor modelling, QUDT represents quantities and units, and OWL-Time represents temporal concepts [? ? ? ?]. Recent reviews of ontology-driven building energy management show that Brick, SAREF, SSN/SOSA, and energy-domain ontologies can support building data standardisation, operational analytics, fault detection, demand response, and building digital twins [? ? ?]. However, these vocabularies mainly operate at the building, device, sensor, or system level. They do not directly express the planning relationships needed for residential electrification: which dwelling archetypes are served by which MV/LV transformers, how heat-pump and EV adoption change coincident catchment demand, how LV impacts propagate to MV and HV assets, and how capacity-risk KPIs trace back to source data and scenario assumptions.

Recent research has also extended semantic integration toward BIM-GIS integration, city-scale digital twins, and urban information models. CityGML and EnergyADE provide structured representations for 3D city models and urban energy applications [30, 31]. BIM-GIS integration studies show how semantic models can bridge building models, geospatial data, and urban-scale information systems [? ? ? 32?]. Ontology-based city information models further show how linked data can connect BIM, GIS, IoT, and asset-management data for digital-twin cities [32, 33]. These studies are relevant because residential energy digital twins also require spatial integration across buildings, administrative boundaries, weather zones,

and infrastructure assets. However, the main contribution of this literature lies in urban information integration, not in transformer-level capacity-risk assessment. Most BIM-GIS and city digital-twin workflows do not represent residential technology-transition scenarios, do not attach hourly archetype demand profiles to LV transformer catchments, and do not derive utilisation or overload indicators for distribution-network assets.

Energy-system ontologies and grid vocabularies cover complementary parts of the problem. The Common Information Model (CIM) represents electricity-network assets, topology, and operational data [34]. Grid and energy ontologies, including SARAGON and related semantic grid models, describe network entities, automation concepts, asset relationships, and energy-system interoperability [35? ?]. GeoSPARQL supports spatial representation and spatial querying [36], while PROV-O supports provenance and lineage tracking [37]. SHACL provides a graph validation language for checking RDF data against shape constraints [38]. Together, these vocabularies and standards can describe buildings, devices, measurements, spatial objects, network assets, units, time, validation constraints, and provenance. However, existing vocabularies do not provide a single queryable structure that connects residential archetype demand, technology-adoption scenarios, LV transformer catchments, upstream network hierarchy, capacity indicators, schema validation, and investment-grade provenance. This missing connection matters because distribution planners need asset-level risk intelligence, not only building-level semantics or aggregate demand forecasts.

Table 1: Selected ontology and digital-twin studies relevant to residential electrification planning.

Study	Main focus	What it includes	What it does not include	Relevance to NEDT
Reinhart and Cerezo Davila [3]	Urban building energy modelling	Defines UBEM as a method for simulating building-stock demand at urban scale.	Does not connect demand outputs to transformer catchments or network capacity.	Supports the demand-modelling foundation but exposes the spatial mismatch with grid planning.
Swan and Ugursal [7]; Kavgić et al. [8]	Residential stock energy modelling	Reviews bottom-up and top-down approaches for residential energy demand.	Does not provide semantic integration with LV grid assets.	Supports the need for archetype-based stock modelling.
Pauwels and Terkaj [18]	Semantic BIM / ifcOWL	Translates IFC into OWL and enables linked-data representation of BIM.	Does not model residential stock scenarios or transformer capacity.	Provides a foundation for semantic building data representation.
Rasmussen et al. [21]	Building topology ontology	Represents sites, buildings, storeys, spaces, zones, and elements.	Does not represent energy-transition scenarios or network assets.	Provides reusable building-topology concepts for semantic digital twins.
Balaji et al. [25]	Smart-building metadata	Represents building equipment, sensors, points, and relationships.	Does not address national stock modelling or LV capacity planning.	Useful for device and operational metadata, but insufficient for transformer-level planning.
Daniele et al. [27]; Poveda-Villalón et al. [39]	Smart appliances and building devices	Represents devices, functions, energy services, and building systems.	Does not connect dwellings to MV/LV transformers or model overload risk.	Provides reusable device and energy semantics.
Shi et al. [32]; Malek et al. [40]	BIM-GIS and city digital twins	Integrates building, GIS, and urban-scale information.	Does not evaluate transformer-level utilisation under HP and EV adoption.	Supports the spatial-integration logic needed by NEDT.
Uslar et al. [34]; Tun et al. [35]	Grid ontologies and network asset semantics	Represents electricity-network assets, topology, and grid relationships.	Does not represent dwelling archetypes, residential technology adoption, or building-stock demand profiles.	Provides reusable grid-asset semantics for NEDT.
Knublauch and Kontokostas [38]; Lebo et al. [37]	Validation and provenance	Demonstrates competency questions, SHACL validation, and provenance modelling in semantic workflows.	Does not apply these mechanisms to transformer-level electrification risk.	Supports the use of SHACL and PROV-O for auditable NEDT outputs.
This paper	Transformer-level residential electrification planning	Connects dwelling archetypes, demand profiles, HP/EV/PV scenarios, LV transformers, upstream grid assets, capacity KPIs, SHACL validation, and PROV-O provenance.	Does not replace detailed engineering studies or DSO operational power-flow tools.	Fills the gap between building-stock energy modelling, semantic building data, and distribution-network capacity planning.

Based on the literature review, this paper identifies four gaps. First, existing UBE and building-stock models rarely resolve residential demand to MV/LV transformer catchments or combine that demand with asset-capacity data. Second, current workflows do not routinely propagate technology-adoption evidence from building energy records to transformer-catchment scale. Third, existing ontologies do not formally connect dwelling archetypes, technology-transition scenarios, hourly demand profiles, electricity-network assets, and capacity-risk indicators within a single queryable framework. Fourth, current workflows provide limited support for schema validation and auditable provenance suitable for investment justification.

The aim of this paper is to develop and demonstrate a semantic digital-twin framework that connects residential building-stock data, archetype-based hourly demand profiles, technology-transition scenarios, and electricity-network capacity data to support transformer-level electrification planning. Rather than treating residential decarbonisation as only an aggregate demand forecasting problem, the paper focuses on the asset-level question faced by distribution planners: which MV/LV transformers are exposed to additional heat-pump and EV demand, when do they approach capacity limits, and what source data and assumptions explain each risk indicator?

This paper introduces the National Energy Digital Twin (NEDT), a semantic framework for transformer-level residential electrification planning. The novelty lies not in creating another generic building ontology, but in connecting building-stock energy modelling to distribution-network capacity assessment through a queryable and provenance-aware knowledge graph. The NEDT reuses established vocabularies where possible, including BOT, SAREF, SAREF4ENER, SAREF4BLDG, SARAGON, GeoSPARQL, QUDT, OWL-Time, SOSA/SSN, SHACL, and PROV-O, and extends them with the missing concepts required for residential LV planning. These concepts include archetype counts, hourly demand profiles, technology-transition scenarios, scenario operators, dwelling-to-transformer service links, capacity KPIs, overload flags, and provenance records linking outputs to source data and assumptions. The NEDT builds directly on linked data work by Hoare et al. that established a hierarchical knowledge graph connecting Irish residential buildings, small areas, and electricity network assets through the DDIM server. Where that prior work created a descriptive utility-aware graph linking buildings to transmission nodes and LV transformers, the NEDT formalises four additional capabilities: an OWL 2 DL T-Box for machine-readable vocabulary reuse; SHACL constraints for A-Box schema validation; PROV-O classes and relations for computational lineage; and a scenario vocabulary for expressing technology-adoption assumptions. The implementation provides a semantic integration and early-stage planning framework that complements detailed operational digital-twin deployments.

The paper makes five contributions. First, it defines an OWL 2 DL ontology extension that connects residential archetypes, technology-transition scenarios, demand profiles,

LV transformers, upstream network assets, and capacity-risk indicators. Second, it presents a national-scale knowledge-graph workflow that ingests and harmonises building, address, census, energy-performance, demand-profile, and network-capacity data, and validates the resulting A-Box using SHACL. Third, it develops a building-to-LV enrichment method that links georeferenced dwellings to candidate serving transformers by nearest-station assignment, and records the assignment process through PROV-O. Fourth, it implements a scenario-evaluation layer that propagates heat-pump, EV, and rooftop PV adoption assumptions from dwelling archetypes to transformer-level hourly demand and utilisation indicators, with each catchment’s response determined by its own archetype composition. Fifth, it demonstrates the framework on Ireland with a reproducible 2024 residential-demand baseline and documents the additional data and validation needed before scenario outputs can support operational LV, MV, or HV decisions.

The remainder of the paper is organised as follows. Section 2 presents the NEDT methodology, including the implementation architecture, data sources, ontology design, knowledge-graph construction, SHACL validation, building-to-LV enrichment, scenario simulation, and overload indicators. Section 3 reports knowledge-graph coverage, baseline network capacity, Scenario 1 impacts, and competency-question performance. Section 4 discusses planning implications, generalisability, and limitations. Section 5 concludes and identifies future work.

2. Methodology

The NEDT implements a semantic workflow that links national residential building data, archetype-based demand profiles, technology-transition scenarios, and electricity-network capacity within a knowledge-graph framework. Figure 1 summarises the five-stage workflow. First, the workflow ingests and harmonises national datasets. Second, the NEDT ontology materialises cleaned source records as RDF triples and validates them using SHACL shapes. Third, the workflow infers cross-dataset relationships that the raw inputs do not contain, most importantly the linkage between each dwelling and its candidate serving LV transformer. Fourth, the computational layer evaluates archetype-based hourly demand profiles under baseline and scenario conditions. Fifth, the workflow derives transformer utilisation and overload indicators by comparing scenario-specific coincident peak demand with available transformer capacity.

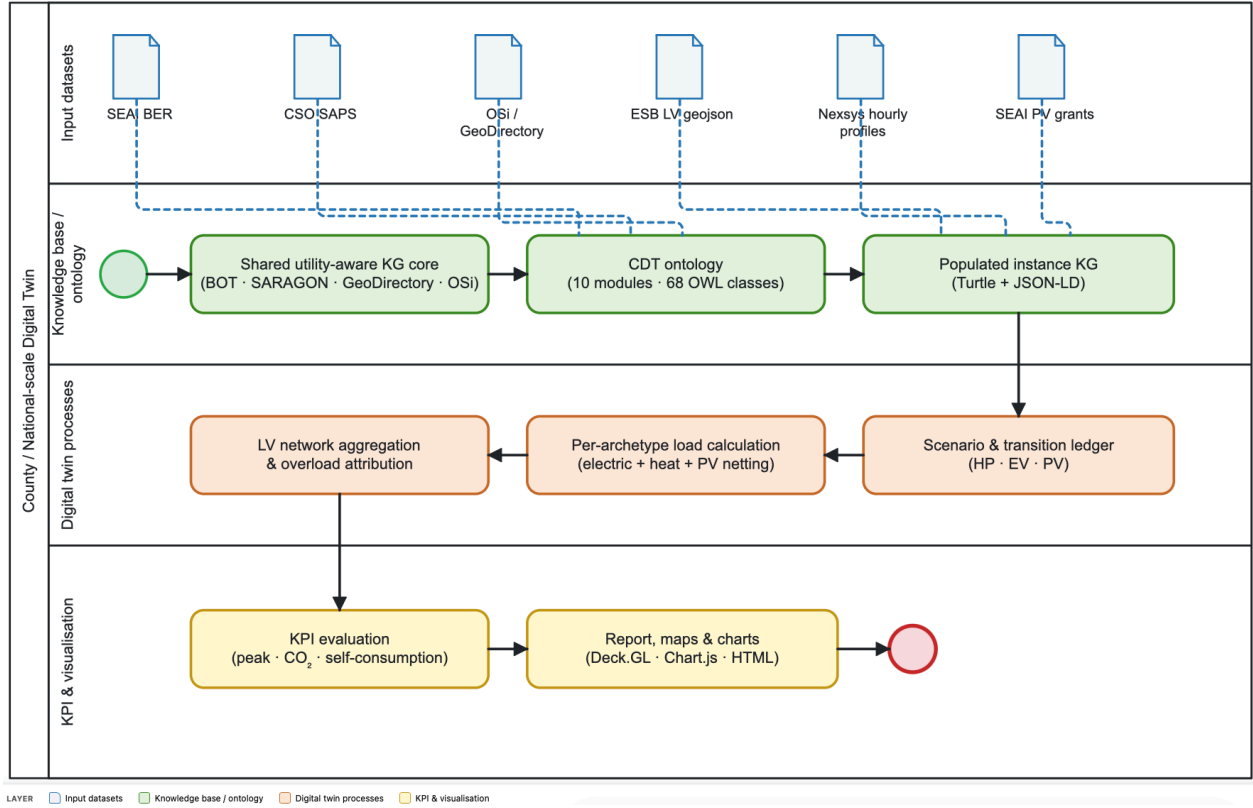


Figure 1: Overarching NEDT methodology. Grey elements show source datasets and competency questions elicited from stakeholders. Orange elements show the main contributions: the NEDT ontology extension, SHACL validation layer, and PROV-O-traceable KPI layer.

2.1. Implementation Architecture

The NEDT uses a two-layer pipeline. The *semantic layer* contains the OWL 2 DL T-Box ontology, SHACL validation shapes, and an Apache Jena Fuseki SPARQL 1.1 triple-store. This layer defines the schema, specifies competency questions as SPARQL query patterns, constrains A-Box instances before the workflow releases results, and records provenance through PROV-O activities. The *computational layer* uses a pandas and GeoPandas pipeline operating on CSV and GeoJSON serialisations of the A-Box. This layer executes demand aggregation, scenario transitions, utilisation calculations, and cross-scale roll-up operations.

The two-layer design separates schema complexity from execution performance. The semantic layer acts as the authoritative layer for schema validation and competency-question specification. The computational layer provides the execution engine for numerical planning outputs. Key derived outputs, including demand profiles, utilisation KPIs and capacity

flags, are written back into the triplestore as populated instance triples so that they become queryable through SPARQL. The released Scenario 1 result graph materialises one `nedt:CapacityKPI` per evaluated LV station and links each KPI to its station, scenario and generating PROV-O activity. Section 3.4 reports the corresponding SHACL validation and executed competency-question outputs.

The NEDT T-Box is not tied to a single graph store backend. The current implementation stores the graph in Apache Jena Fuseki with SPARQL 1.1, which was selected to support native SHACL validation and standards-compliant SPARQL execution for competency-question evaluation. The same OWL 2 DL T-Box can alternatively be deployed using the Neo4j and NeoSemantic combination already demonstrated in related DDIM work [41], in which NeoSemantic ingests OWL and RDFS vocabularies and maps classes, properties, and individuals to Neo4j nodes, relationships, and labels. That deployment has been applied at national scale in Ireland, linking the residential building stock to over 46,000 LV transformers [41], and offers a larger developer community, simpler Cypher query syntax, and native LLM and RAG integration suited to operational DSO environments. Every relationship in the Neo4j property graph maps 1:1 to an RDF triple: for example, `(b:Building)-[:NEAREST_LV_STATION]->(lv:LVStation)` is the Cypher form of `nedt:isServedBy` `ex:b` `ex:lv`, so the semantic content is identical across both backends and queries can be ported between Cypher and SPARQL without loss of meaning. Adopters may therefore select whichever backend fits their infrastructure without modifying the T-Box or SHACL shape graph.

2.2. Data Sources

Six national datasets provide the main A-Box inputs. Table 2 summarises their roles and principal limitations.

Table 2: Primary data sources used in the NEDT workflow.

Dataset	Role in NEDT	Main limitation
GeoDirectory GeoAddress and BuildingIntelligence	Provides the national address spine, geocoded dwelling centroids, and building attributes used for LV assignment.	Reports floor area in coarse bands; estimates construction period rather than surveying it directly.
SEAI BER Research Tool	Provides dwelling energy-performance attributes, heating system type, and archetype classification inputs.	Covers approximately 60% of the national stock and is biased toward recently sold or retrofitted dwellings.
CSO Small Area Population Statistics	Provides Small Area dwelling counts, household structure, occupancy, and heating-fuel distributions for stock expansion and calibration.	Releases data at aggregate Small Area level; does not provide individual dwelling attributes.
ESB Networks capacity and network data	Provides LV station locations, capacity information, and parent-network relationships used for utilisation analysis.	May not fully expose derating, operational switching states, or exact nameplate values for all assets.
OSi administrative boundaries	Provides county, local authority, and Small Area geometries for spatial containment and hierarchical aggregation.	Administrative geographies do not necessarily align with electrical catchment boundaries.
CSO measured-energy series	Provides independent county-level electricity consumption used as a calibration and validation signal for aggregate demand.	Is too aggregated to validate individual transformer loads directly.

2.3. *Ontology Design and Semantic Scope*

The NEDT specifies the ontology as an OWL 2 DL T-Box using a reuse-first strategy. The released artefact includes syntax parsing, graph construction, and SHACL-validation tests. The T-Box uses subsumption, domain and range axioms, property inverses, one functional property and one transitive property; it does not require the more expressive OWL 2 constructs, so the DL classification reflects the profile it validates against rather than a claim of high expressivity. The ontology adopts existing vocabularies where they provide stable domain coverage: BOT for building topology, SAREF and SAREF4ENER for device and energy concepts, SARAGON for electricity-network assets, GeoSPARQL for spatial representation, QUDT for quantities and units, OWL-Time for temporal concepts, SOSA/SSN

for observations, and PROV-O for provenance.

The NEDT extension introduces the classes and properties required to connect residential building-stock modelling with distribution-network planning. Core NEDT classes include `nedt:Archetype`, `nedt:ArchetypeCount`, `nedt:DemandProfile`, `nedt:Scenario`, `nedt:ScenarioOperator`, `nedt:TechnologyTransition`, `nedt:LVStation`, and `nedt:CapacityKPI`. The key connecting relationship is `nedt:isServedBy` (domain: `nedt:Dwelling`; range: `nedt:LVStation`), which links each dwelling to its candidate supplying transformer. Roll-up across the network hierarchy is expressed through a transitive super-property. `nedt:hasParentMVFeeder` and `nedt:hasParentHVSubstation` are both sub-properties of the transitive `nedt:hasUpstreamAsset`, whose inverse `nedt:feeds` is aligned in intent with the SARAGON `feeds` relation. An OWL 2 reasoner therefore entails the full $LV \rightarrow MV \rightarrow HV$ closure from the two published one-step links, so demand can be aggregated at any tier without materialising the closure in the A-Box. This provides a machine-interpretable network-hierarchy pattern for downstream reasoning and roll-up queries. The released HermiT regression test loads the T-Box, checks that no class is unsatisfiable, and verifies the inferred two-step LV–MV–HV relation from a minimal three-asset fixture. This test establishes the semantic mechanism; it is not a claim that the current case-study data has complete or verified connectivity.

The ontology aligns external vocabularies using `rdfs:subClassOf`, SKOS mapping relations, and `rdfs:seeAlso`. At instance level the released graph contains no `owl:sameAs` assertions. Identity across the source datasets is resolved deterministically during extraction, on normalised station names and address identifiers, and a single harmonised IRI is minted for the resolved entity, so no cross-graph co-reference statement is required. We note this because `owl:sameAs` is often assumed in linked-data integration architectures; here the resolution happens before RDF materialisation rather than being expressed in the graph. Appendices Appendix A and Appendix B provide the full class and property inventories.

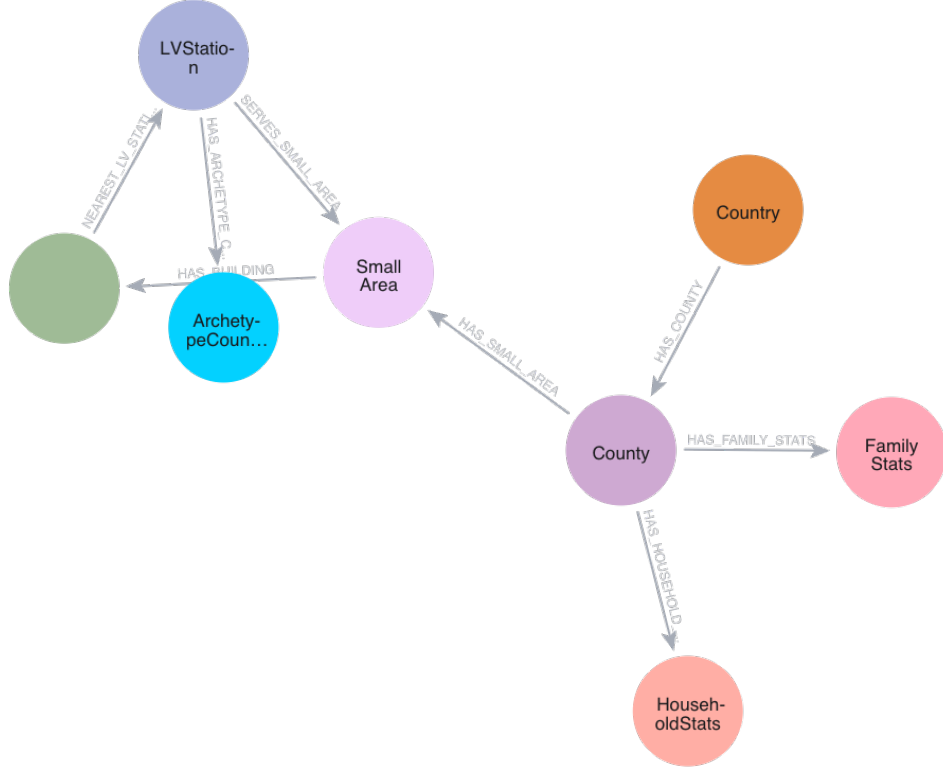


Figure 2: Neo4j schema of the NEDT knowledge graph showing eight node labels (Country, County, SmallArea, Building, LVStation, ArchetypeCounts, FamilyStats, HouseholdStats) and eight relationship types. Every relationship maps 1:1 to an RDF triple — for example, (b:Building)-[:NEAREST_LV_STATION]->(lv:LVStation) corresponds directly to `ex:b nedt:isServedBy ex:lv` — making the graph queryable through both Cypher and SPARQL without schema modification.

2.3.1. Competency Questions

Fifteen competency questions frame the evaluation. They were formulated from the planning concerns of DSO, TSO, SEAI, local-authority and DECC stakeholders as encountered in project engagement and in the published network and policy documents of those bodies; they are not the output of a structured elicitation exercise with recorded participants and protocol, and no such study is claimed here. Section 3.4 reports which of the fifteen were executed against the released A-Box. CQ1–CQ2 assess archetype and building-stock representation. CQ3–CQ4 assess scenario parameters and technology transitions. CQ5–CQ7 assess LV station demand, overload, and flexibility indicators. CQ8–CQ10 assess county, national, and network-level KPIs. CQ11–CQ15 assess sensitivity, provenance, dwelling assignment, MV/HV roll-up, and validation observations. Appendix E provides SPARQL

skeletons for all fifteen questions.

2.4. Knowledge Graph Construction and Validation

The workflow constructs the NEDT graph by mapping cleaned source records into RDF triples that instantiate the ontology. The T-Box contains class definitions, property definitions, alignment axioms, and SHACL shapes. The A-Box contains populated instances derived from the source datasets. The workflow retains source identifiers where possible and mints harmonised NEDT IRIs for integrated entities. Dataset-specific RDF mappers transform tabular data, while GeoJSON-to-RDF transformations convert spatial data. The workflow stores the graph in Apache Jena Fuseki through the DDIM server and maintains the T-Box and A-Box as separate named graphs to support schema stability and incremental updates.

The workflow applies SHACL validation before releasing any results. The shape graph enforces required links, expected datatypes, and controlled-vocabulary constraints. The workflow accepts a graph state only when `pySHACL` reports `conforms = true` against the full shape graph. The validation covers both a Dublin county graph and the full national Ireland graph. Table 3 reports the results.

Table 3: `pySHACL` validation results at county and national scope. Both scopes apply the same five node shapes and eleven property constraints.

Scope	Triples	Stations	Archetypes	Stock nodes	Violations	Elapsed (s)
Dublin	627,419	6,094	540	73,451	0	62.7
Ireland	3,330,755	36,234	741	388,606	0	347.0

Two qualifications are necessary for these results to be read correctly.

First, the **stock nodes** column counts `nedt:ArchetypeCount` individuals, each representing a (station, archetype) cohort, not individual dwellings. The A-Box materialises 388,606 such cohort nodes covering 2,136,436 dwellings; it does not instantiate 2.1 million `nedt:Dwelling` individuals. Queries that enumerate stock at this scope therefore operate over cohorts, and CQ13, which retrieves individual dwellings for a station, is not answerable against this serialisation.

Second, the shape graph reported here was generated by the same script that serialised the A-Box. A conformance result obtained this way tests serialisation consistency, not data quality, and zero violations under such a configuration carries little evidential weight. To address this, the released shape graph has been rewritten to constrain value ranges, controlled vocabularies and cross-field consistency rather than only the presence of triples that the extraction always emits, and it is distributed with a fixture of nine deliberately malformed

instances together with a self-test asserting that all nine are rejected. A shape graph that conforms against that fixture is treated as broken. Appendix Appendix D reproduces the shapes; the Artefact and Data Availability section gives the repository.

Subject to those qualifications, the validation returned zero violations at both scopes, confirming that every instance satisfies the required cardinality, datatype and controlled-vocabulary constraints. These validation results apply to the schema layer of the A-Box. The pandas pipeline computes capacity KPI values — utilisation ratios, overload flags, and derived indicators — and loads them into the graph as a subsequent step, after which CQ5–CQ10 return populated result sets.

PROV-O is used to represent graph-construction, enrichment and simulation activities, each recording source datasets, generated artefacts, processing stage and timestamp. In the released Scenario 1 result graph, every materialised `nedt:CapacityKPI` carries a `prov:wasGeneratedBy` link to the executable scenario-workflow activity; CQ12 verifies this path directly. Provenance coverage for the legacy baseline A-Box remains partial, so provenance claims in this paper are scoped to the released Scenario 1 KPI write-back rather than to every historical analysis artefact.

2.5. Building-to-LV Transformer Enrichment

The NEDT must link each dwelling to a candidate serving LV transformer. Let $\mathcal{B}$ denote the set of dwellings, where each $b \in \mathcal{B}$ has a geocoded centroid $\mathbf{p}_b$ in EPSG:2157, and let $\mathcal{T}$ denote the set of LV transformers, where each $t \in \mathcal{T}$ has a location $\mathbf{p}_t$ and, where available, a service-area polygon $\mathcal{A}_t$. The assignment rule selects the candidate serving transformer as

$$t^*(b) = \begin{cases} t \in \mathcal{T} : \mathbf{p}_b \in \mathcal{A}_t, & \text{if a service-area polygon is available,} \\ \arg \min_{t \in \mathcal{T}} \|\mathbf{p}_b - \mathbf{p}_t\|_2, & \text{otherwise.} \end{cases} \quad (1)$$

In the implementation reported here the second branch is the one that runs. ESB Networks does not publish per-station service-area polygons, so no independent polygon layer is available: the catchment polygons used for aggregation and mapping are concave or convex hulls, or $\sqrt{\text{kVA}}$ -weighted Voronoi cells, constructed *from* the nearest-transformer assignment itself. Using them as an assignment rule would therefore be circular, and they are not used for that purpose. All dwelling-to-transformer links in this paper are nearest-transformer assignments, and the first branch of Equation (1) is retained only to show how a published connectivity layer would enter the method if a DSO released one. The workflow stores all geometries in Irish Transverse Mercator (EPSG:2157), whose planar distortion across the state remains below 10 ppm, so Euclidean distances provide a faithful proxy for geodesic distances.

The workflow materialises each assignment as $\langle b, \text{nedt:isServedBy}, t^*(b) \rangle$ and attaches a PROV-O activity that records the enrichment method, source data snapshots, and timestamp.

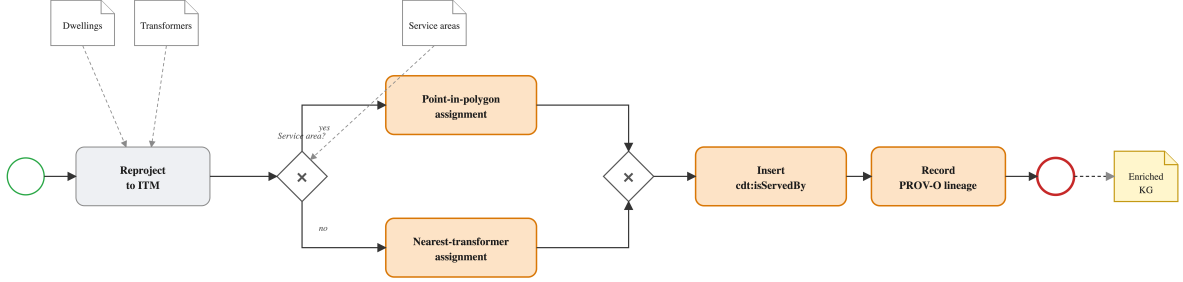


Figure 3: Building-to-LV enrichment process. Dwellings are linked to candidate LV transformers using published service-area polygons where available and nearest-transformer assignment otherwise. The PROV-O activity records the assignment method, source snapshots, and timestamp.

The assessment evaluates assignment accuracy using two metrics. The *dwelling-level accuracy* α measures the fraction of dwellings for which the assigned transformer matches a DSO-supplied ground-truth record. The *catchment-level Jaccard similarity* J_τ measures the ratio of the intersection to the union of the predicted and ground-truth dwelling sets for each transformer τ , averaged across transformers. A preliminary assessment on the Dundalk (Co. Louth) public pilot feeders gives $\alpha \approx 0.87$; dwellings on catchment boundaries explain most misassignments. This figure should be treated as indicative only. It derives from a single town on urban and peri-urban feeders, the sample size is small, and the underlying comparison has not been released as a reproducible artefact. The catchment-level Jaccard similarity J_τ is defined here because it is the appropriate metric for aggregate catchment agreement, but it has not yet been computed, and no confidence interval is available for α . Dundalk is also the favourable case: rural feeders follow road corridors that diverge most from Euclidean proximity, and it is precisely the rural oil-heated stock that dominates the heat-pump conversion cohort. No claim of national assignment accuracy is made in this paper. Full ground-truth validation against a national DSO connectivity sample is the first item of future work in Section 5.

2.6. Archetype Demand Simulation and Scenario Transitions

The simulation stage assigns hourly energy demand profiles to dwellings through archetype simulations. The workflow classifies each archetype by dwelling type, BER energy-performance class, occupancy category, heating-fuel type, and weather zone. Let $a(b)$ denote the archetype of dwelling b . Each archetype has a precomputed 8760-hour electricity and heat profile derived from building-energy simulation using EnergyPlus and decomposed into end-uses using

the occupancy-driven method of [42]. These profiles represent typical dwelling behaviour for each archetype class. Section 4.4 discusses variation within archetype classes.

For archetype a and scenario S , the scenario-specific electricity demand at hour h is

$$d_{a,S}(h) = d_{a,0}(h) + \phi_{a,S}^{\text{HP}} \Delta d_a^{\text{HP}}(h) + \phi_{a,S}^{\text{EV}} \Delta d_a^{\text{EV}}(h) - \phi_{a,S}^{\text{PV}} d_a^{\text{PV}}(h), \quad (2)$$

where $d_{a,0}(h)$ is the baseline electricity profile, $\Delta d_a^{\text{HP}}(h)$ is the additional electric load associated with heat-pump conversion, $\Delta d_a^{\text{EV}}(h)$ is the EV charging profile, and $d_a^{\text{PV}}(h)$ is the rooftop PV generation. The terms $\phi_{a,S}^{\text{HP}}$, $\phi_{a,S}^{\text{EV}}$, and $\phi_{a,S}^{\text{PV}}$ represent scenario adoption fractions. The graph stores these fractions explicitly and links them to resulting demand profiles via `nedt:evaluatedUnderScenario`. Appendix Appendix C provides the Scenario 1 parameterisation.

Rooftop-PV construction and netting.. The PV term is generated from the NEDT Nexsys uncontrolled-charging workbook, rather than inferred from EV demand. For each valid archetype sheet, the hourly PV `generation` series is divided by its own peak; the resulting curves are averaged, zero-padded to 8760 hours where necessary, and multiplied by 1000/1284 to represent a 1,000 kWh/kWp/year Irish irradiance yield. The common normalised shape is then scaled by a 2.5 kWp system for terraced dwellings, 3.5 kWp for semi-detached dwellings, and 4.0 kWp for detached dwellings and bungalows; apartments are excluded. At each hour, self-consumed PV reduces electricity demand only to zero, while the configured export component is retained as net injection. This preserves the distinction between behind-the-meter consumption and exported generation when the station-level series is formed.

For each transformer t , hourly demand is the sum of the archetype profiles of the dwellings assigned to it,

$$D_{t,S}(h) = \sum_{a \in \mathcal{A}_t} n_{a,t} d_{a,S}(h), \quad (3)$$

where $n_{a,t}$ is the number of dwellings of archetype a in catchment t , and the coincident peak is

$$P_{t,S} = \max_{h \in \{1, \dots, 8760\}} D_{t,S}(h). \quad (4)$$

Two properties of this operator matter for what follows. First, it is *allocate-then-maximise*: profiles are summed within the catchment and the annual maximum is taken on that sum. The alternative, taking a county-level maximum and distributing it by dwelling count, is not equivalent, because it makes every station in a county share one demand per dwelling and reduces station utilisation to a function of the dwellings-per-kVA ratio. Under Equation (3) the archetype composition of each catchment, and therefore BER class, occupancy and heating fuel, determines station demand: the 37,287 modelled stations take 15,868 distinct values of peak demand per dwelling, with a coefficient of variation of 0.41.

Second, no separate coincidence factor is applied. The quantity $\max_h \sum_a n_{a,t} d_a(h)$ is already an after-diversity maximum, because temporal diversity between archetypes is expressed in the summation itself. Multiplying it by a further after-diversity factor would discount diversity twice. The resulting national mean of 1.20 kW per dwelling, and 1.46 kW per dwelling for the Dublin stock, sit close to the 1.5 kW after-diversity maximum demand used in Irish and UK LV design practice, which provides an external check that the diversity treatment is dimensionally correct.

Roll-up to MV and HV follows the same rule. The coincident peak of a parent asset is the annual maximum of the summed hourly series of its children, not the sum of their individual peaks. Across 467 MV feeders the coincident peak is 0.988 of the sum of child LV peaks (IQR 0.953–0.995), so inter-catchment diversity is real but modest at this voltage level. All voltage tiers in Section 3 derive from the same station-level series, so utilisation values are comparable across tiers.

2.6.1. Archetype profile coverage

The archetype-count file distinguishes 744 archetype signatures, while the simulated profile library holds 300, structured as five dwelling types $\times$ three heating systems $\times$ BER class $\times$ occupancy categories 1–6. Heat-pump archetypes exist only at BER A–B and fossil-fuelled archetypes only at BER B–E. Signatures without an exact profile are served by the nearest available profile in the same dwelling-type and heating-system cell, minimising distance in BER and then occupancy. Under this rule 75.7% of dwellings match an archetype exactly and 24.3% are served by a neighbour. Electrically heated dwellings (139,368) and the 98 solar-thermal dwellings have no corresponding profile and are mapped to the heat-pump and gas archetypes respectively. These substitutions are recorded per record and are a limitation of the profile library rather than of the method; Section 4.4 discusses their effect.

2.6.2. Demand validation

Aggregating the station-level series of Equation (3) to county level gives an independent check against CSO metered residential electricity consumption for 2024. No calibration factor is applied: the comparison is between raw simulated demand and measured consumption. Table 4 reports the result. Modelled national consumption is 8,477 GWh against 8,943 GWh metered, a normalised mean bias error of -5.2% , with a mean absolute percentage error of 11.8% and a CV(RMSE) of 11.2% across the 26 counties, and a coefficient of determination of 0.9935. These are descriptive errors over 26 annual spatial aggregates. They are not monthly time-series calibration statistics and must not be compared with ASHRAE Guideline 14 monthly-calibration thresholds. The comparison therefore checks only approximate aggregate-volume consistency; it does not validate hourly peaks, transformer assignment, or capacity classifications.

This replaces the county calibration used in earlier versions of this workflow, in which modelled electricity was multiplied by county-specific factors ranging from 0.745 to 1.444 so that totals reconciled with CSO. Those factors are not used here. The residual county errors reported in Table 4 range from -19.8% to $+23.6\%$ and are largest in counties with sparse BER coverage, which indicates that the remaining error is a property of the stock characterisation rather than of the demand operator.

Table 4: County-level validation of modelled residential electricity against CSO metered consumption, 2024. No calibration factor is applied.

Indicator	Value	Interpretation	Met	
Counties compared	26	—	—	
Modelled national total	8,477 GWh	—	—	
Metered national total (CSO)	8,943 GWh	—	—	
NMBE	-5.2%	annual aggregate error	—	No
MAPE	11.8%	—	—	
CV(RMSE)	11.2%	annual aggregate error	—	
R^2	0.9935	dominated by county size	—	
Largest positive error	Donegal, $+23.6\%$	—	—	
Largest negative error	Monaghan, -19.8%	—	—	

ASHRAE compliance claim is made: these are annual county totals, not monthly calibration observations.

2.7. Capacity and Overload Indicators

The following indicator is a *screening proxy*, not a transformer thermal-loading measurement. It becomes an engineering loading assessment only when real and reactive power, temperature/derating, switching state, and a validated service connection are available. Consequently, threshold counts in this paper identify candidates for DSO study rather than verified overloads.

Given the coincident peak demand $P_{t,S}$ from Equation (4), the workflow calculates transformer utilisation as

$$U_{t,S} = \frac{P_{t,S}/\text{pf}}{C_t k_\theta}, \quad (5)$$

where C_t is the rated capacity from the ESB Networks asset register, pf is an effective load power factor converting real power to apparent power, and $k_\theta = 1 + 0.005(\theta_{\text{rated}} - \theta_{\text{amb}})$ is the IEC 60076-2 ambient correction, clamped to $[0.85, 1.20]$. The implementation computes pf as a load-weighted mean (pf = 0.95 base, 0.87 heat pump, 0.97 EV). **For the results reported in Section 3, both corrections are inactive (pf = 1, k_θ = 1), so utilisation is computed as real power in kW divided by rated apparent power in kVA.** Because

a transformer thermal rating is a kVA constraint, this understates utilisation by a factor of 1/pf, approximately 5% at baseline and more under heat-pump uptake, where inverter drives lower the effective power factor. We state this explicitly rather than leave it implicit in the units.

The workflow classifies a transformer as *over-capacity* when $U_{t,s} > 1.0$ and as *near-capacity* when $0.9 \leq U_{t,s} \leq 1.0$. For MV and HV assets, the workflow aggregates LV-level demands through the parent-feeder hierarchy.

Capacity imputation. Rated capacity is unavailable for 954 of the 37,287 stations (2.6%). These stations are assigned a default of 200 kVA. Only 16 of the 3,009 over-capacity stations (0.5%) carry an imputed rating, so the headline counts are not materially driven by the substitution. Because the denominator of Equation (5) is the rated capacity, an imputed rating determines over-capacity status deterministically for the affected assets, whatever their modelled demand. Individual station verdicts for imputed assets are not evidence, but the national totals are insensitive to the choice. The graph flags these stations with `nedt:hasCapacityImputed`, and the SHACL shape graph requires the flag on every station, so that imputed capacities cannot propagate silently into KPIs. Utilisation estimates additionally carry interval uncertainty where the published capacity is banded rather than exact, so comparisons close to the 100% threshold should be interpreted with caution.

3. Results

This section reports four sets of results: knowledge graph coverage and enrichment quality; the 2024 baseline capacity status across LV, MV, and HV tiers; scenario impacts under a compound residential electrification case; and SHACL-conformant competency-question evaluation against the populated national result graph.

3.1. Knowledge Graph Coverage and Enrichment

The constructed knowledge graph contains **2,137,827 nodes** across four entity types: 26 counties, 13,254 Small Areas, 46,377 LV substations, and 2,078,170 georeferenced residential building records. Building and dwelling counts differ throughout this paper and are not interchangeable: a building record may contain more than one dwelling, so the 2,078,170 building records correspond to the 2,136,436 dwellings carried through the demand analysis. Approximately **4.20 million directed edges** encode building-to-Small-Area containment (2,078,170), building-to-LV-station assignment (2,078,170), and station-to-county location links (46,377). Of the 46,377 substations in the graph, 41,309 match at least one residential building. The remaining substations likely serve non-residential or sparsely populated areas.

Three station populations appear in this paper and are not interchangeable. The spatial graph contains 46,377 substations, of which 41,309 have at least one assigned building. The

capacity analysis in Sections 3.2 and 3.3 covers the 37,287 stations that carry a BER-matched archetype demand profile. The serialised A-Box used for SHACL validation and competency-question benchmarking (Sections 2.4 and 3.4) contains 36,234 stations, being the 37,287 less 1,053 whose county could not be resolved and which were therefore excluded from the RDF export.

The building-to-transformer assignment has a median distance of **134.6 m** and a mean distance of **435.7 m**. The enrichment process assigns 65.2% of buildings within 200 m, 85.5% within 1 km, and 95.4% within 2 km. A small tail of 3,068 buildings, equivalent to 0.15%, have assignment distances above 5 km, concentrated in sparse western rural areas. Urban counties show shorter assignment distances, while rural western counties show longer distances, consistent with lower transformer density in those regions. These distance statistics describe spatial proximity; they do not directly measure assignment accuracy. The preliminary dwelling-level accuracy from the Dundalk pilot ($\alpha \approx 0.87$) gives the best available accuracy estimate. Future work will complete full national ground-truth validation.

3.2. Baseline Network Capacity

The 2024 baseline represents as-is residential demand before the workflow applies any additional HP, EV, or rooftop PV adoption. National hourly electricity demand shows little seasonal trend, though it cycles daily between roughly 0.2 and 2.8 GWh/h, while heat demand is strongly seasonal with winter peaks near 9 GWh/h.

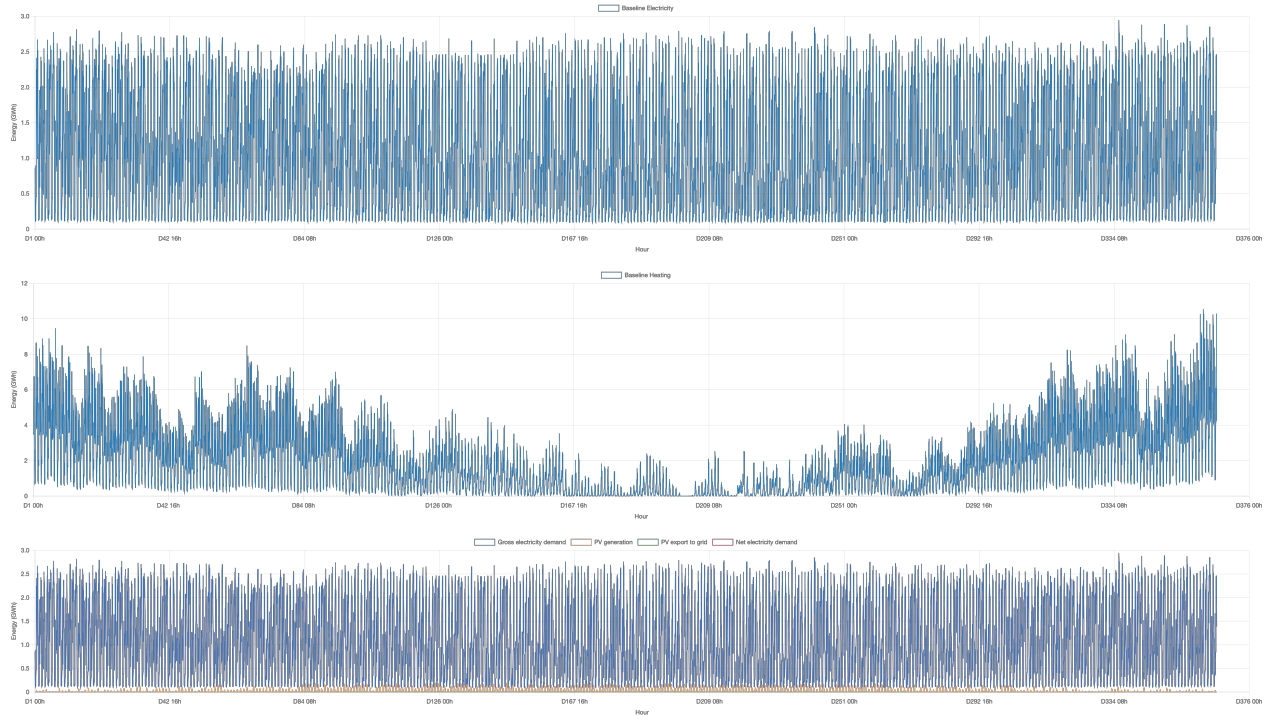


Figure 4: Baseline 2024 national hourly profiles. *Top*: electricity demand. *Middle*: aggregated fossil heat demand. *Bottom*: rooftop PV generation.

The baseline model covers 37,287 MV/LV substation catchments with BER-matched residential demand profiles and serves 2,136,436 dwellings. These LV catchments roll up to 496 MV feeders and 100 HV substations (Table 5).

Table 5: Network scope of the 2024 baseline by voltage tier.

Voltage tier	Assets modelled	Dwellings served	Source
LV, MV/LV substations	37,287	2,136,436	ESB Networks + BER archetype file
MV, 38 kV feeders	496	–	ESB parent linkage
HV, 110/220 kV nodes	100	–	Upstream nodes with TSO interface

Table 6 summarises baseline utilisation across all three voltage tiers. At LV level, 3,009 stations, or 8.1%, already exceed rated capacity and serve 304,878 dwellings. A further 609 LV stations fall in the 90–100% near-capacity band. At MV level, 4 of 467 feeders exceed rated capacity.

Peak demand per dwelling is no longer uniform across the estate. It has a national mean of 1.20 kW and a median of 1.227 kW, with a 5th–95th percentile range of 0.671–1.820 kW and 15,868 distinct values. It also rises systematically with transformer size, from a mean of 1.03 kW per dwelling on units rated at or below 50 kVA to 1.47 kW on units above 400 kVA, reflecting the larger and less efficient dwellings typically served by larger transformers. Dublin, with its higher apartment and gas-heating share, has a mean of 1.46 kW per dwelling against the national 1.20 kW. All HV substations remain below 90% utilisation.

Table 6: As-is 2024 capacity status by voltage tier. *Near-capacity* = 90–100% utilisation. All tiers derive from the same station-level hourly series of Equation (3); MV and HV values are coincident roll-ups, not independently parameterised runs. The Dublin row is the subset of stations in that county, shown because the archetype mix there differs markedly from the national stock.

Voltage tier	Total	Red >100%	Near-cap.	Stressed	OK	Peak util.
LV, national	37,287	3,009	609	9.7%	33,669	997%
LV, Dublin area	6,348	118	54	2.7%	6,176	553%
MV, 38 kV	467	4	2	1.3%	461	144%
HV, 110/220 kV	108	0	0	0.0%	108	—

The LV stress pattern varies strongly across space. Figure 5 shows that urban cores — Dublin, Cork, Limerick, and Galway — generally have lower stress, reflecting higher substation density and a more recently reinforced network. Rural and peri-urban catchments show higher concentrations of over-capacity and near-capacity stations. In these areas, smaller legacy transformers, predominantly 50 kVA units installed between the 1970s and 1990s, now serve catchments whose dwelling counts have grown substantially since the original network construction.

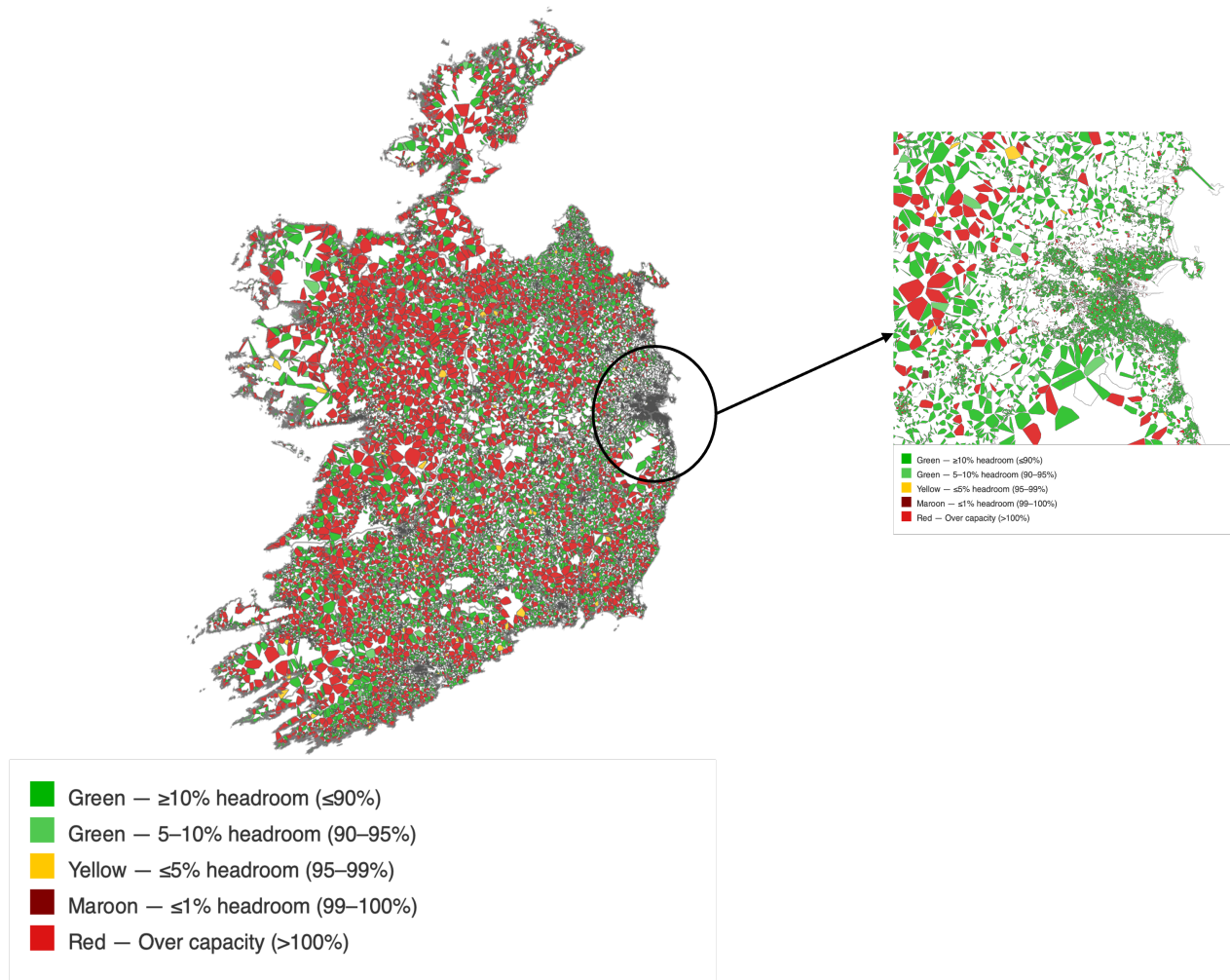


Figure 5: LV station utilisation under the 2024 baseline. Each polygon represents the Voronoi service area of one LV station, coloured by peak utilisation band. Grey outlines show county boundaries.

Table 7 ranks the highest-utilisation LV transformers nationally. Of the 3,009 over-capacity stations, 79.3% have a 50 kVA rating, the smallest standard unit in the ESB Networks plant. These transformers were sized for loads dominated by electric lighting and resistive appliances. Unlike a dwellings-per-kVA screen, this ranking responds to catchment composition: the listed stations span 1.13–1.54 kW per dwelling, so two catchments of similar size can differ by a third in modelled peak according to their archetype mix. Only 16 of the 3,009 over-capacity stations carry an imputed rated capacity.

Table 7: Ten highest-utilisation LV transformers nationally in the 2024 baseline. Station identifiers are anonymised per data-sharing protocol. Excess power is peak demand minus rated capacity. No station in this ranking has an imputed rated capacity.

Station	Bldgs	Rated (kVA)	kW/dwell.	Util. (%)	Excess (kW)
BA***166Z	3,236	400	1.23	996.7	+3,586.8
KI***4037	295	50	1.54	907.5	+403.8
TU***002X	307	50	1.40	862.2	+381.1
PR***419	237	33	1.15	826.6	+239.8
CR***323Y	310	50	1.23	763.8	+331.9
TE***507	323	50	1.13	731.6	+315.8
BR***4060	306	50	1.14	698.1	+299.1
HO***265	273	50	1.23	672.4	+286.2
DO***4103	267	50	1.23	656.6	+278.3
FA***4180	568	100	1.15	651.0	+551.0

At MV level, Table 8 lists the four over-capacity 38 kV feeders under the coincident roll-up. These differ from the feeders identified in earlier versions of this workflow, which computed MV demand on a separate basis from LV; the values below are the annual maximum of the summed hourly series of each feeder’s child LV stations, so they are consistent with Table 6 by construction.

Table 8: MV (38 kV) feeders exceeding rated capacity in the 2024 baseline, as coincident roll-ups of the per-catchment LV series.

Feeder	Peak (kW)	Rated (kVA)	Util. (%)
Portarlinton	7,205	5,000	144.1
Turlough Road	6,490	5,000	129.8
Kinsale	11,680	10,000	116.8
Buttevant	4,248	4,000	106.2

3.3. Electrification Scenario Impacts

Scenario 1 is an illustrative planning case, designed to demonstrate how technology-transition assumptions propagate through the NEDT information model. It is not intended as a forecast of realised adoption or a substitute for asset-specific network studies.

Scenario 1 represents a near-term compound adoption case that combines heat-pump, EV, and rooftop PV uptake. Appendix Appendix C provides the full parameterisation. The scenario converts 20% of eligible BER C–E oil-heated and gas-heated dwellings to heat

pumps. It sets EV penetration at 10% of the national private-car stock and rooftop PV penetration at 10% of eligible non-apartment dwellings.

Scenario 1 converts 20% of eligible BER C–E oil- and gas-heated dwellings to heat pumps, a national total of 210,843 conversions, equips 243,265 dwellings with EV chargers, and applies rooftop PV to 183,108 eligible non-apartment dwellings. Of the 168,459 incremental EV assignments, 6,844 (4.1%) are allocated to heat-pump-transition cohorts; HP–EV co-adoption is therefore represented explicitly rather than excluded. Table 9 shows that detached oil-boiler dwellings form the largest HP adoption cohort. This reflects Ireland’s rural housing stock outside the gas network.

Table 9: Eight largest heat-pump adoption cohorts by building type and incumbent fuel under Scenario 1, aggregated over BER and occupancy classes. Smaller bungalow, duplex, temporary-structure and terraced-house cohorts are retained in the machine-readable transition ledger.

Building type	Incumbent fuel	HP units
Detached	Oil boiler	67,129
SemiDetached	Gas boiler	46,023
Terraced	Gas boiler	33,089
SemiDetached	Oil boiler	29,242
Apartment	Gas boiler	14,579
Detached	Gas boiler	10,573
Terraced	Oil boiler	8,487
Apartment	Oil boiler	1,644

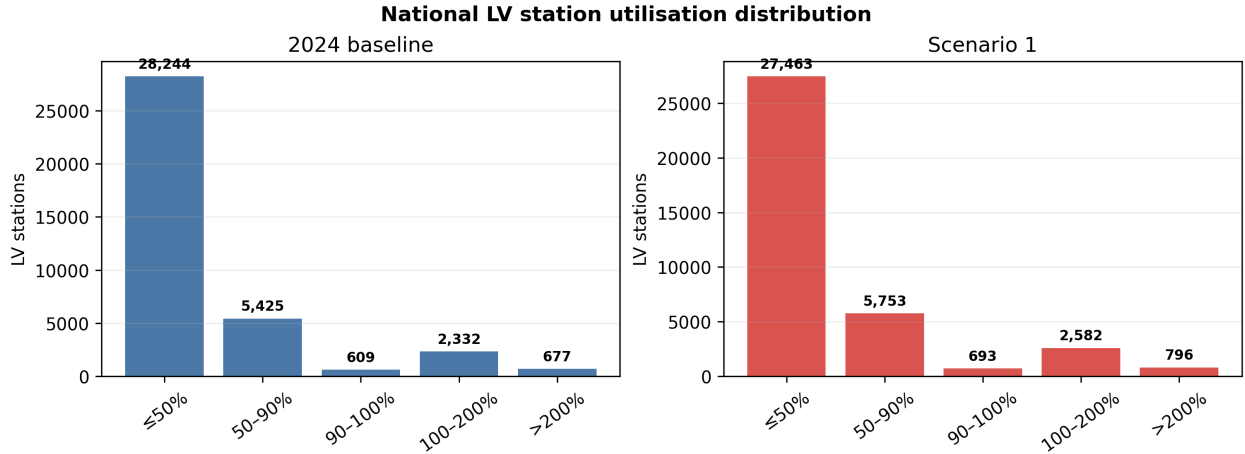


Figure 6: National LV utilisation distribution under the regenerated Scenario 1 workflow (HP + EV + PV per Appendix Appendix C), compared with the 2024 baseline.

Figure 6 shows LV utilisation under Scenario 1. Because baseline and scenario are computed from the same operator and the same profile library, the two are directly comparable. Table 10 reports the national LV result.

Table 10: National LV capacity status under Scenario 1 compared with the 2024 baseline, both computed with Equation (3). *Near-capacity* = 90–100% utilisation.

Indicator	Baseline	Scenario 1	Change
Stations modelled	37,287	37,287	—
Over-capacity (>100%)	3,009	3,378	+369
Near-capacity (90–100%)	609	693	+84
Stressed (%)	9.70%	10.92%	+1.2 pp
Dwellings in over-capacity catchments	304,878	343,973	+39,095
95th-percentile utilisation	128.1%	136.8%	+8.7 pp
99th-percentile utilisation	246.2%	264.9%	+18.7 pp
Mean peak per dwelling	1.203 kW	1.275 kW	+6.0%
<i>Origin of the 401 newly over-capacity stations</i>			
from near-capacity band (90–100%)	—	245	61.1%
from below 90% utilisation	—	156	38.9%

Asset and connectivity quality screen.. The workflow retains all stations in the national model-screened statistics, but distinguishes a reconciliation cohort from interpretable screening candidates. An asset enters this cohort when its modelled utilisation exceeds 500% or its archetype input represents more than 1,000 dwellings. The screen flags 46 stations (14,115 modelled dwellings; 1.36% of the 3,378 red flags). They are retained for data-quality prioritisation, not presented as routine reinforcement candidates, and are reported separately in Table 11 and the machine-readable `asset_reconciliation_exceptions.csv` artefact.

The highest modelled value, 1,154.6%, is an input-reconciliation exception, not a headline planning result. For BA***166Z, the archetype input assigns 3,236 dwellings whereas the independent building-to-station assignment file contains 1,868 dwellings (a ratio of 1.73). The published availability snapshot lists 400 kVA installed capacity and 78 kVA allocated load (19.5% utilisation). That snapshot is not time-aligned overload ground truth, but the discrepancy precludes interpreting the modelled 4.62 MW peak as an observed overload. Across the 46 records, 20 have an archetype-to-assignment dwelling ratio outside $\pm 20\%$ and 16 have an availability utilisation at or below 20%. The executable workflow therefore materialises this count reconciliation diagnostic for every station. The all-station distribution is reported with robust 95th and 99th percentiles in Table 10; its maximum is not used to characterise national planning exposure.

The scenario response is heterogeneous, which is the substantive difference from a uniformly scaled projection. The per-station ratio of scenario to baseline peak has a mean of 1.058 and a median of 1.000, but ranges from 0.989 at the 5th percentile to 1.261 at the 95th, across 3,779 distinct values after rounding to four decimal places. Stations therefore change rank under the scenario, and the assets that move into the over-capacity band are those whose catchments actually contain the oil-heated and gas-heated stock being converted, rather than those that happened to be closest to the threshold. The small number of stations with a ratio below one are catchments whose baseline electric-vehicle share already exceeded the flat 10% penetration the scenario imposes.

Of the 401 stations that newly exceed capacity, 245 (61.1%) were in the 90–100% near-capacity band at baseline and 156 (38.9%) were below it. The near-capacity band therefore captures rather more than half of the newly stressed population but misses a substantial minority, which is an argument for screening on modelled scenario utilisation directly rather than on baseline headroom alone.

Table 11 separates the reconciliation cohort from lower-extreme model-screened stations. The latter remain conditional candidates for detailed network study; they are not validated overloads.

Table 11: Scenario 1 asset-reconciliation exceptions, shown separately from planning-ranked candidates. “Modelled dwellings” is the archetype-input count used by the operator. The complete 46-record audit is supplied as a machine-readable artefact.

Cohort	Stations	Modelled dwellings	Scenario utilisation	Interpretation
BA***166Z	1	3,236	1,154.6%	Input/asset reconciliation required
Other audit cohort	45	10,879	500.5–906.3%	Input/asset reconciliation required
Lower-extreme red flags	3,332	329,858	100.0–498.9%	Conditional screening candidates

The present reproducibility bundle reports Scenario 1 capacity indicators at LV level. The baseline MV roll-up is retained in Section 3.2; scenario-specific MV/HV indicators are not reported until the parent-asset identifiers are reconciled across the station connectivity and published capacity registers.

3.4. Competency Question Evaluation

The Scenario 1 result graph contains 764,115 triples when loaded with the NEDT T-Box and national stock graph. It conforms to the distributed SHACL shape graph. Four representative competency questions were executed directly against this graph using RDFLib SPARQL: scenario parameters (CQ3), county archetype inventory (CQ1), over-capacity stations (CQ6), and provenance trace (CQ12). This compact set exercises scenario semantics, stock coverage, station-level planning outputs, and traceability. CQ3 resolves the adoption-fraction datatype properties of `nedt:Scenario`; CQ1 follows `nedt:inCounty` from `nedt:ArchetypeCount`; CQ6 joins `nedt:CapacityKPI`, `nedt:forLVStation`, and the SOSA utilisation result; and CQ12 follows `prov:wasGeneratedBy`. Together these queries demonstrate the ontology path from stock evidence and scenario assumptions to a traceable LV screening result.

Table 12: Executed competency-question outputs against the SHACL-conformant Scenario 1 result graph (764,115 triples). Times are single local RDFLib SPARQL executions and are reported for reproducibility rather than as a triplestore performance benchmark.

Query / competency question	Time (ms)	Rows
Scenario parameters (CQ3)	86.9	1
County archetype inventory (CQ1)	99.5	26
Over-capacity stations (CQ6)	2,393.6	3,378
Provenance trace (CQ12)	3.3	10

CQ6 returns 3,378 Scenario 1 over-capacity stations, which agrees with Table 10; this is the planning result rather than an empty query placeholder. CQ12 retrieves the activity associated with each materialised KPI, enabling a user to trace a station-level result back to the Scenario 1 workflow execution. The scripts and machine-readable query output are supplied with the artefact.

3.5. Interpretation of published network availability data

The ESB Networks availability extract is used as an asset-register input and as a reconciliation signal, not as validation data for the residential-demand model. Its station fields express installed capacity, capacity allocated to connections, and available headroom; they are not timestamped measurements of transformer power flow. A modelled coincident residential peak and a committed-capacity value therefore answer different questions. In particular, zero available capacity can reflect committed connections, non-residential demand, network-planning policy, or operational constraints rather than an observed residential overload.

For this reason, the reproducibility bundle does not report a correlation between modelled peak demand and published allocation values as a validation metric. Instead, it retains the

availability fields in the `asset_reconciliation_exceptions.csv` artefact to support the asset-record review described in Section 3.3. Station-level demand validation requires time-aligned DSO metering and connectivity data; until then, all utilisation results are interpreted as conditional infrastructure-exposure indicators.

4. Discussion

4.1. Planning Implications

The results show that residential electrification creates a transformer-level planning problem, not only a national demand-growth problem. Of the 401 stations that newly exceed capacity under Scenario 1, 245 (61.1%) were in the 90–100% band at baseline and 156 (38.9%) were below it (Table 10). Baseline headroom is therefore an incomplete screen on its own. Because the scenario response varies by catchment, from 0.989 to 1.261 at the 5th and 95th percentiles, a station well below the amber band but with a heavily oil-heated catchment can overtake one sitting just under it. The practical implication is that the watchlist should be built from modelled scenario utilisation rather than from baseline headroom, which is only possible once station demand responds to catchment composition. Either way, planners cannot wait for measured overloads in the most vulnerable catchments given reinforcement lead times of three to seven years.

The digital twin provides a screening layer for this task. It ranks LV stations by baseline utilisation and scenario-driven overload, helping planners prioritise detailed studies on the small subset of assets most likely to require intervention. Spatial clustering in the ranked output also supports bundled reinforcement programmes, which typically cost less per unit of capacity than isolated asset replacements. Driver attribution is available at station level, because each catchment’s scenario response is generated from its own archetype composition. A station whose ratio is 1.261 is one whose catchment holds a high share of the oil-heated stock being converted; a station near 1.0 is one that does not. This distinction matters because each driver implies a different intervention, and it was not identifiable under a uniform allocation. Planners should nonetheless treat NEDT outputs as a targeted exposure screen for validation and intervention planning, not as final engineering designs, and should prefer published headroom where it exists (Section 3.5).

4.2. The Knowledge Graph as Enabling Infrastructure

The knowledge graph provides the integration mechanism that makes the planning outputs possible. Each source dataset answers only part of the planning question. Network data contain capacity but not dwelling demand. BER data describe dwellings but not their serving transformers. Demand profiles describe archetypes but do not attach those archetypes to real network assets. The graph connects these layers through explicit relationships between

dwellings, archetypes, demand profiles, LV transformers, upstream assets, and scenarios. This structure makes it possible to ask which transformer serves which dwelling mix, under which scenario, and with which dominant load driver.

This structure also supports explainability. A future operational deployment can trace a flagged station back to its archetype-cohort distribution and scenario assumptions once derived KPIs are materialised with their PROV-O activities. The released A-Box does not yet provide that end-to-end trace: it materialises stock as archetype cohorts rather than dwelling individuals, and it omits derived capacity KPIs (Section 2.4). SHACL validation reduces the risk that missing capacities, invalid links, or inconsistent records propagate silently into KPIs. PROV-O records preserve lineage from derived indicators to source datasets and processing steps, which supports capital-investment justification under regulatory planning frameworks.

4.3. Generalisability to Other Jurisdictions

The workflow requires four core data layers: a georeferenced building or address register, a building energy database, archetype demand profiles, and LV transformer asset data with locations and capacities. Parent-network links and measured electricity consumption data improve roll-up and calibration, but a minimum deployment does not strictly require them.

Table 13: Data layer mapping from Ireland to the United Kingdom.

Layer	Ireland	UK equivalent	Main adaptation
Building and address	GeoDirectory; OSi	GeoPlace UPRN	Address and identifier mapping
Building energy data	SEAI BER Register	Energy Performance of Buildings database	BER/SAP archetype crosswalk
Demand profiles	Irish archetype simulations	BREDEM, TM54, NEED, or local simulations	Weather and profile recalibration
Network assets	ESB Networks data	DNO open data; RIIO-ED2 datasets	DNO-specific schema mapping
Spatial allocation	Dwelling-to-transformer assignment	UPRN, MPAN, or DNO connectivity data	Replace proximity heuristic where direct connectivity is available

The main transfer challenge is institutional. Ireland has a single DSO, while the UK has multiple DNOs with different data formats and access arrangements. A federated knowledge-graph deployment can address this challenge by allowing each organisation to retain its own instance data while mapping assets to a shared ontology vocabulary.

The DDIM framework provides a complementary management layer for deploying the NEDT ontology in new jurisdictions. The current DDIM implementation uses Neo4j with the NeoSemantic plugin and exposes a forms-based, no-code interface through which domain experts can upload datasets, define owl:sameAs equivalences and owl:part-of hierarchies between columns, and construct a core context spine without requiring SPARQL or graph-query

expertise. This approach has been validated at national scale in Ireland, linking residential buildings to LV transformers. For transfer to a new jurisdiction, a planner needs only to substitute the equivalent national datasets for example, GeoPlace UPRN address records and EPC data for the UK and map them to the shared NEDT vocabulary using the same form-based workflow, without modifying the underlying ontology or server code.

4.4. Limitations

Within-archetype variability. Station demand now responds to catchment composition, but all dwellings within an archetype still receive an identical profile. Real homes within one archetype differ in insulation, heating schedule, occupancy and appliance mix. This matters most for catchments of 10–30 dwellings, where a few atypical households can shift the coincident peak substantially, and it is the likely reason that inter-catchment diversity at MV is small: the coincident MV peak is 0.988 of the sum of child LV peaks, whereas measured networks show more. A stochastic profile layer would be the natural extension.

Archetype profile coverage and substitution. As set out in Section 2.6.1, 75.7% of dwellings match a simulated archetype exactly and 24.3% are served by the nearest available profile. Electrically heated dwellings, 6.5% of the stock, have no corresponding profile and are represented by heat-pump archetypes, which understates their peak because resistive heating is roughly three times less efficient. Results for catchments dominated by electrically heated stock should be treated as lower bounds.

The scenario bundles a fabric retrofit with the heat pump. The profile library carries heat-pump archetypes only at BER A–B and fossil-fuelled archetypes only at BER B–E. A BER D oil-heated dwelling therefore has no BER D heat-pump counterpart and converts to a BER B heat-pump archetype, so every modelled conversion is implicitly also a deep fabric retrofit. Peak demand still rises on conversion, by a factor of 2.68 for a detached BER D oil dwelling, so the direction of the result holds, but Scenario 1 is not a pure electrification scenario and understates the network impact of a heat pump installed into unimproved fabric. Quantifying that gap requires heat-pump profiles at the lower BER classes.

Comparability of the external reference. The ESB Networks figure used in Section 3.5 is allocated rather than measured capacity, and 40% of stations are censored at 0% or 100%. It bounds the model’s exposure but cannot validate its load prediction. A metered comparison against a DSO monitoring sample remains the outstanding validation requirement at station level.

BER coverage and spatial bias. The BER register covers approximately 60% of the national dwelling stock and over-represents recently sold or retrofitted dwellings. It under-represents some rural oil-heated dwellings, which represent the most important HP adoption candidates. This biases the archetype mix that shapes the county aggregate, though it is a smaller effect on the capacity results than the diversity double-count described above.

Transformer assignment accuracy. The building-to-transformer assignment uses spatial proximity and service-area polygons. In rural areas, distribution feeders follow road corridors that do not necessarily correspond to Euclidean proximity. The Dundalk figure ($\alpha \approx 0.87$) is indicative only: single town, small sample, no released artefact, no confidence interval, and the companion Jaccard metric J_τ has not been computed. Because Equation (4) makes station peak a function of assigned dwelling count alone, assignment error propagates directly and proportionally into utilisation. Any assignment conserves aggregate county demand, but individual station verdicts near the threshold are sensitive to it.

Capacity data quality and imputation. Rated capacity is imputed at 200 kVA for 954 stations (2.6%), as described in Section 2.7. Utilisation verdicts for those assets are determined by the imputed value and are not evidence. The national counts are not highly sensitive to the default, and only 16 of the 3,009 over-capacity stations carry an imputed rating, but individual verdicts for those assets are not evidence.

Network asset data quality. Utilisation estimates depend on the accuracy of published transformer ratings. If those ratings are outdated, derated in practice, or inconsistent with switching arrangements, the model inherits those errors. A higher-confidence deployment would require direct integration with the DSO asset-management system.

5. Conclusions and Future Work

This paper presents the National Energy Digital Twin (NEDT), a semantic framework for transformer-level residential electrification planning. The core contribution is an OWL 2 DL ontology that connects building-stock archetypes, hourly demand profiles, technology-transition scenarios, electricity-network assets, capacity-risk indicators, and provenance within a single queryable structure.

The ontology fills a gap between existing building and grid vocabularies. BOT, Brick, and SAREF describe buildings, systems, and devices, while CIM and SARAGON describe electricity-network assets. However, these vocabularies do not express the relationships needed for proactive LV planning: which archetypes are served by which transformers, how HP and EV adoption changes catchment-level demand, how LV results roll up to MV and HV assets, and how derived KPIs trace back to source records and scenario assumptions. The NEDT extension introduces these missing concepts while reusing established ontologies.

The Irish case study represents 2,136,436 dwellings as archetype cohorts associated with 37,287 candidate MV/LV station catchments. The 2024 baseline identifies 3,009 over-capacity and 609 near-capacity LV stations. Modelled annual county electricity has a normalised mean bias of -5.2% and a CV(RMSE) of 11.2% against CSO totals without calibration; this is an aggregate consistency check, not temporal or station-level validation. Scenario 1 illustrates how the ontology can represent technology transitions and associate scenario identifiers with capacity-screening outputs. The resulting indicators support prioritisation of candidate assets for detailed assessment under alternative adoption assumptions. The results show that the framework can organise fragmented building and network data into a transformer-level, scenario-aware screening layer. It provides a structured basis for detailed DSO studies, uncertainty analysis, and progressive KPI materialisation as additional operational data become available.

The ESB Networks availability extract provides a useful asset-register input and a reconciliation signal, but it is not time-aligned transformer metering. The paper therefore does not treat it as validation of modelled peak demand. Its value in the present workflow is to make data-quality exceptions explicit and traceable alongside the scenario KPI, rather than to convert a screening output into an observed-overload claim.

Future work will address five areas, in priority order.

First, station-level validation against metered data. The comparison in Section 3.5 is bounded by its reference, which is committed rather than measured capacity. A monitoring sample from the DSO would allow the demand model to be assessed directly at the asset level, which the county-level validation in Table 4 cannot do.

Second, national DSO connectivity records should validate the dwelling-to-transformer assignment, replacing the indicative Dundalk figure with a national accuracy estimate and a computed catchment-level Jaccard similarity. Because station demand is now built from the dwellings assigned to each catchment, assignment error propagates directly into the archetype mix and therefore into the peak.

Third, the profile library should be extended. Heat-pump archetypes at BER C–E would remove the implicit retrofit bundling described in Section 4.4, and profiles for electrically heated dwellings would remove the largest remaining substitution. A stochastic within-archetype layer would also allow inter-catchment diversity to be represented, which the present deterministic profiles understate.

Fourth, the graph should represent uncertainty explicitly, including BER coverage, profile substitution, transformer rating intervals and imputed capacities. The released Scenario 1 KPI write-back is queryable, but the remaining eleven competency questions should be executed against an expanded operational A-Box, including baseline and uncertainty representations.

Fifth, a second jurisdiction, such as the UK, should test how well the ontology and work-

flow transfer across different building-energy databases, address systems, and DNO schemas.

Artefact and Data Availability

The NEDT ontology (T-Box), the SHACL shape graph, and its deliberately invalid SHACL fixture are available at <https://github.com/buildinginformaticslab/nedt-ontology> under version 1.4.0. `validate_shapes.py` runs the shape graph against `test_shapes_trap.ttl`, a fixture of nine deliberately malformed instances, and fails if any is accepted. `test_ entailment.py` loads the T-Box with HermiT, checks logical consistency, and verifies the transitive LV–MV–HV roll-up fixture. The release also provides `build_kg.py` and `populate_v11_instances.py`. Together they regenerate the base knowledge graph, the example A-Box, and the merged graph; all three conform to the distributed 24-node-shape, 66-property-constraint SHACL suite. The T-Box currently uses the namespace <https://example.org/nedt#>; migration to a permanent w3id.org PURL is pending and the namespace should be treated as provisional until that registration completes.

The distributed `rerun_percatchment.py` regenerates the 2024 baseline per-station profiles, the utilisation-ratio comparison against the published ESB allocation ratio, and the MV coincident roll-up. `scenario1_percatchment.py` regenerates the Scenario 1 station table from the raw station archetype counts, EnergyPlus profile library, national car-stock target, and Nexsys PV workbook. `export_scenario_results_to_rdf.py` materialises the Scenario 1 station output as `scenario1_results.ttl`; `run_competency_queries.py` executes the four reported queries and writes a machine-readable result summary. These scripts, the Scenario 1 result table, reconciliation-exception audit, manifest, and RDF graph are supplied as the reproducibility bundle.

The source datasets are third-party and subject to their own licences: GeoDirectory (commercial licence, Ordnance Survey Ireland and An Post), SEAI BER Research Tool (registration required), CSO Small Area Population Statistics and metered electricity series (open), OSi administrative boundaries (open), and ESB Networks capacity heat-map data (published subject to ESB Networks terms). Station identifiers are anonymised throughout in accordance with the data-sharing protocol, and no dwelling-level record is redistributed. The archetype profile library is derived from EnergyPlus simulations described in Section 2.6.

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Appendix A. Ontology Class Inventory

Table A.14: Class inventory for the NEDT residential energy-infrastructure ontology.

Class	Source ontology	Status	Description	Role in NEDT
Building	BOT	Reused	Physical building entity.	Spatial container for dwellings.
Dwelling	BOT + NEDT	Extended	Residential unit within a building.	Core unit for archetype assignment and transformer linkage.
County	NEDT	Custom	Administrative territorial unit.	County-level aggregation and reporting.
SmallArea	NEDT / OSi / CSO	Custom	Census Small Area spatial unit.	Links stock to SAPS attributes.
LVStation	NEDT + SARAGON	Extended	MV/LV transformer or local LV substation.	Connects dwelling demand to network capacity.
MVFeeder	SARAGON / NEDT	Extended	Medium-voltage parent network asset.	Supports LV-to-MV roll-up.
HVSubstation	SARAGON / NEDT	Extended	High-voltage upstream substation.	Supports MV-to-HV roll-up.
Archetype	NEDT	Custom	Representative dwelling-stock type.	Assigns demand profiles to cohorts.
ArchetypeCount	NEDT	Custom	Count of dwellings assigned to an archetype.	Stock weighting in each spatial unit.
ArchetypeState	NEDT	Custom	Baseline or scenario-specific archetype condition.	Supports technology-state transitions.
HeatingSystem	SAREF / Brick	Reused	Generic heating technology.	Links dwellings and archetypes to heating systems.

Class	Source ontology	Status	Description	Role in NEDT
GasBoiler	SAREF / Brick	Reused	Gas-fired heating technology.	Baseline fossil-heating state.
OilBoiler	SAREF / Brick	Reused	Oil-fired heating technology.	Rural fossil-heating state.
HeatPump	SAREF / Brick	Reused	Electric heat-pump technology.	Electrified heating scenario state.
EVCharger	SAREF / Brick	Reused	Electric-vehicle charging device.	EV charging demand representation.
V1HCharger	NEDT	Custom	One-directional smart charger.	Controlled EV charging.
V2HCharger	NEDT	Custom	Bidirectional vehicle-to-home charger.	Flexible EV operation.
OccupancyModel	NEDT	Custom	Model used to generate occupancy profiles.	Behavioural assumptions for simulation.
OccupancyProfile	NEDT	Custom	Time-varying occupancy representation.	Occupancy-sensitive demand profiles.
Scenario	NEDT	Custom	Baseline or intervention case.	Groups technology assumptions and outputs.
ScenarioOperator	NEDT	Custom	Action that modifies stock or technology state.	HP, EV, and PV scenario logic.
TechnologyTransition	NEDT	Custom	Transition from one technology state to another.	Technology-state changes per scenario.
DemandProfile	NEDT	Custom	Hourly or structured demand profile.	Baseline and scenario demand.
CapacityKPI	NEDT + QUDT	Custom	Capacity-risk indicator.	Utilisation, overload, and headroom.

Class	Source ontology	Status	Description	Role in NEDT
ValidationObservation	NEDT + SOSA	Custom	Observation used to validate model outputs.	Links simulated outputs to measured data.
CountyDemandAggregate	NEDT	Custom	County-level demand summary.	County KPI outputs.
NationalDemandAggregate	NEDT	Custom	National demand summary.	Ireland-level KPI outputs.
TechnologyState	NEDT	Custom	Current technology configuration.	Baseline and scenario technology state.

Appendix B. Ontology Property Inventory

Table B.15: Core object properties for the NEDT ontology.

Object property	Domain	Range	Role in NEDT
hasDwelling	Building	Dwelling	Links buildings to residential units.
locatedInCounty	Building / Dwelling / LVStation	County	County-level aggregation and filtering.
locatedInSmallArea	Dwelling	SmallArea	Links dwellings to SAPS attributes.
isServedBy	Dwelling	LVStation	Links each dwelling to its assigned LV transformer.
hasParentMVFeeder	LVStation	MVFeeder	Supports LV-to-MV roll-up.
hasParentHVSubstation	MVFeeder	HVSubstation	Supports MV-to-HV roll-up.
hasServiceArea	LVStation	geo:Feature	Links a transformer to a service-area polygon.
instantiatesArchetype	Dwelling	Archetype	Assigns a dwelling to an archetype.
usesHeatingSystem	Dwelling / Archetype	HeatingSystem	Represents heating technology.

Object property	Domain	Range	Role in NEDT
hasEVCharger	Dwelling	EVCharger	Represents EV charging capability.
evaluatedUnderScenario	DemandProfile / KPI / ArchetypeState	Scenario	Links outputs to the scenario under which they were produced.
appliesScenarioOperator	Scenario	ScenarioOperator	Links a scenario to intervention operators.
changesTechnologyState	ScenarioOperator	TechnologyTransition	Represents a technology change.
forArchetype	DemandProfile	Archetype	Links a demand profile to its archetype.
forLVStation	CapacityKPI	LVStation	Links a KPI to an LV asset.
aggregatesDemandFrom	CountyDemandAggregate / NationalDemandAggregate	DemandProfile / AggregationNode	Supports multi-scale roll-up.
validatedAgainstObservation	DemandProfile / KPI	ValidationObservation	Links simulated outputs to validation data.
prov:wasGeneratedBy	KPI / DemandProfile / Assignment	prov:Activity	Links outputs to their generating process.
prov:used	prov:Activity	Dataset / ScenarioAssumption	Records input datasets and assumptions used.

Appendix C. Scenario 1 Parameterisation

Table C.16: Scenario 1 parameterisation by technology and charging mode. EV Tariff 3/4/Dynamic percentages are fractions of the controlled sub-fleet; Uncontrolled is a fraction of the full EV fleet. V1H and V2H have no uncontrolled mode.

Technology	Parameter	Value	Basis
Heat pump	Oil-heated dwellings converting to HP	20%	Scenario assumption
Heat pump	Gas-heated dwellings converting to HP	20%	Scenario assumption
EV	Dwelling penetration	10%	Scenario assumption
EV	Charging mode — Uncontrolled	50%	of EV fleet
EV	Charging mode — Tariff 3 (ToU)	80%	of controlled fleet
EV	Charging mode — Tariff 4	10%	of controlled fleet
EV	Charging mode — Dynamic	10%	of controlled fleet
V1H	Penetration	25%	of EV fleet
V1H	Charging mode — Tariff 3 (ToU)	80%	of V1H fleet
V1H	Charging mode — Tariff 4	10%	of V1H fleet
V1H	Charging mode — Dynamic	10%	of V1H fleet
V2H	Penetration	25%	of EV fleet
V2H	Charging mode — Tariff 3 (ToU)	80%	of V2H fleet
V2H	Charging mode — Tariff 4	10%	of V2H fleet
V2H	Charging mode — Dynamic	10%	of V2H fleet
Rooftop PV	Eligible non-apartment dwelling penetration	10%	Scenario assumption
Rooftop PV	Share with export capability	20%	of PV-equipped dwellings
Rooftop PV	Gross generation exported to grid	30%	of exporters' output
Rooftop PV	System size: terraced	2.5 kWp	roof-area norm
Rooftop PV	System size: semi-detached	3.5 kWp	roof-area norm
Rooftop PV	System size: detached / bungalow	4.0 kWp	roof-area norm
Rooftop PV	Annual-yield normalisation	1,000 kWh/kWp	Nexsys shape $\times 1000/1284$

HP–EV co-adoption is permitted. After heat-pump transitions, the EV target is allocated across all remaining no-EV archetype components, including heat-pump-transition cohorts. Scenario 1 therefore assigns 6,844 of the 168,459 incremental EVs (4.1%) to a heat-pump-transition cohort. This allocation is deterministic and recorded in the Scenario 1 transition

ledger.

Appendix D. SHACL Validation Shapes

The shape graph constrains value ranges, controlled vocabularies and cross-field consistency, not only the presence of triples that the extraction always emits. Representative shapes are reproduced below; the full graph contains 24 node shapes and 66 property constraints.

The graph is distributed with `test_shapes_trap.ttl`, a fixture of nine deliberately malformed instances, and `validate_shapes.py`, which asserts that all nine are rejected. A shape graph that reports `conforms = true` against that fixture is treated as broken and the self-test exits non-zero. This guards against the failure mode in which a shape graph and the A-Box it validates are generated by the same code and conformance becomes tautological.

```
@prefix sh:    <http://www.w3.org/ns/shacl#> .
@prefix rdfs:  <http://www.w3.org/2000/01/rdf-schema#> .
@prefix prov:  <http://www.w3.org/ns/prov#> .
@prefix xsd:   <http://www.w3.org/2001/XMLSchema#> .
@prefix nedt:  <https://example.org/nedt#> .

nedt:DwellingShape
  a sh:NodeShape ;
  sh:targetClass nedt:Dwelling ;
  sh:property [ sh:path nedt:isServedBy ;
                sh:class nedt:LVStation ;
                sh:minCount 1 ; sh:maxCount 1 ;
                sh:message "Exactly one serving LV station." ] ;
  sh:property [ sh:path nedt:hasAssignmentDistanceM ;
                sh:datatype xsd:decimal ;
                sh:minInclusive 0 ; sh:maxInclusive 20000 ;
                sh:message "Assignment distance outside [0,20000] m
                           indicates a failed spatial join." ] .

nedt:LVStationShape
  a sh:NodeShape ;
  sh:targetClass nedt:LVStation ;
  sh:property [ sh:path nedt:hasCapacityValue ;
                sh:datatype xsd:decimal ;
                sh:minCount 1 ; sh:maxCount 1 ;
                sh:minExclusive 0 ; sh:maxInclusive 2000 ] ;
```

```

    sh:property [ sh:path nedt:hasCapacityImputed ;
                  sh:datatype xsd:boolean ; sh:minCount 1 ;
                  sh:message "Silent capacity imputation is not permitted." ] .

nedt:ArchetypeShape
  a sh:NodeShape ;
  sh:targetClass nedt:Archetype ;
  sh:property [ sh:path nedt:hasBuildType ; sh:minCount 1 ;
                sh:in ( "Detached" "SemiD" "Terraced"
                        "Apartment" "Bungalow" ) ] ;
  sh:property [ sh:path nedt:hasBERCategory ; sh:minCount 1 ;
                sh:in ( "A" "B" "C" "D" "E" "F" "G" ) ] .

nedt:CapacityKPIShape
  a sh:NodeShape ;
  sh:targetClass nedt:CapacityKPI ;
  sh:property [ sh:path nedt:evaluatedUnderScenario ;
                sh:class nedt:Scenario ; sh:minCount 1 ; sh:maxCount 1 ;
                sh:message "Unscoped KPIs cannot be compared." ] ;
  sh:property [ sh:path nedt:hasOverloadHours ;
                sh:datatype xsd:integer ;
                sh:minInclusive 0 ; sh:maxInclusive 8760 ] ;
  sh:property [ sh:path prov:wasGeneratedBy ; sh:minCount 1 ;
                sh:message "Every KPI must trace to a PROV-O activity." ] .

nedt:CountyDemandAggregateShape
  a sh:NodeShape ;
  sh:targetClass nedt:CountyDemandAggregate ;
  sh:property [ sh:path nedt:hasCountyCalibrationFactor ;
                sh:datatype xsd:decimal ; sh:minCount 1 ;
                sh:minInclusive 0.5 ; sh:maxInclusive 2.0 ;
                sh:message "Undeclared calibration would allow scaled
                           output to be reported as validation." ] .

```

Appendix E. SPARQL Skeletons for Competency Questions

The following skeletons define the minimum graph patterns that a conforming NEDT A-Box must support. Production queries include dataset-specific namespaces and additional filters.

Common prefixes.

```
PREFIX nedt: <https://example.org/nedt#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX prov: <http://www.w3.org/ns/prov#>
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
```

CQ1: Dwelling archetypes in county X.

```
SELECT ?archetype ?count
WHERE {
  ?ac a nedt:ArchetypeCount ;
      nedt:forCounty ?county ;
      nedt:instantiatesArchetype ?archetype ;
      nedt:hasCount ?count .
  FILTER (?county = nedt:county_X)
}
ORDER BY DESC(?count)
```

CQ2: Building stock attributes for archetype A.

```
SELECT ?archetype ?buildType ?berClass ?occupancy ?heatingSystem
WHERE {
  ?archetype a nedt:Archetype ;
      nedt:hasBuildType ?buildType ;
      nedt:hasBERCategory ?berClass ;
      nedt:hasOccupancyCategory ?occupancy ;
      nedt:usesHeatingSystem ?heatingSystem .
  FILTER (?archetype = nedt:archetype_A)
}
```

CQ3: Scenario parameters for scenario S.

```
SELECT ?scenario ?hpFraction ?evFraction ?pvFraction ?scenarioYear
WHERE {
  ?scenario a nedt:Scenario ;
      nedt:hasScenarioYear ?scenarioYear ;
      nedt:hasHPAdoptionFraction ?hpFraction ;
      nedt:hasEVAdoptionFraction ?evFraction ;
```

```

        nedt:hasPVAdoptionFraction ?pvFraction .
    FILTER (?scenario = nedt:scenario_S)
}

```

CQ4: Technology transitions in scenario S.

```

SELECT ?transition ?fromArchetype ?toArchetype ?unitCount
WHERE {
    ?op a nedt:ScenarioOperator ;
        nedt:changesTechnologyState ?transition .
    ?transition nedt:fromArchetype ?fromArchetype ;
        nedt:toArchetype ?toArchetype ;
        nedt:transitionCount ?unitCount .
    ?scenario nedt:appliesScenarioOperator ?op .
    FILTER (?scenario = nedt:scenario_S)
}
ORDER BY DESC(?unitCount)

```

CQ5: Archetype breakdown for LV station T.

```

SELECT ?station ?archetype ?count ?scenario
WHERE {
    ?ac a nedt:ArchetypeCount ;
        nedt:forLVStation ?station ;
        nedt:instantiatesArchetype ?archetype ;
        nedt:hasCount ?count ;
        nedt:evaluatedUnderScenario ?scenario .
    FILTER (?station = nedt:station_T)
    FILTER (?scenario = nedt:scenario_S)
}
ORDER BY DESC(?count)

```

CQ6: Over-capacity LV stations under scenario S.

```

SELECT ?asset ?scenario ?utilisation ?peakDemand ?capacity
WHERE {
    ?kpi a nedt:CapacityKPI ;
        nedt:forLVStation ?asset ;
        nedt:evaluatedUnderScenario ?scenario ;
        nedt:hasUtilisation ?utilisation ;
        nedt:hasPeakDemandKW ?peakDemand ;

```

```

        nedt:hasRatedCapacityKVA ?capacity .
    FILTER (?scenario = nedt:scenario_S)
    FILTER (?utilisation > 1.0)
}
ORDER BY DESC(?utilisation)

```

CQ7: Solar PV export.

```

SELECT ?archetype ?scenario ?selfConsumption ?gridExport ?reverseHours
WHERE {
    ?state a nedt:FlexibilityState ;
        nedt:forArchetype ?archetype ;
        nedt:evaluatedUnderScenario ?scenario ;
        nedt:hasSelfConsumptionRatio ?selfConsumption ;
        nedt:hasGridExport ?gridExport ;
        nedt:hasReverseFlowHours ?reverseHours .
    FILTER (?archetype = nedt:archetype_A)
    FILTER (?scenario = nedt:scenario_S)
}

```

CQ8: County peak demand and emissions under scenario S.

```

SELECT ?county ?scenario ?peakMW ?annualEnergy ?annualCO2
WHERE {
    ?agg a nedt:CountyDemandAggregate ;
        nedt:forCounty ?county ;
        nedt:evaluatedUnderScenario ?scenario ;
        nedt:hasPeakDemandMW ?peakMW ;
        nedt:hasAnnualEnergyUse ?annualEnergy ;
        nedt:hasAnnualCO2 ?annualCO2 .
    FILTER (?county = nedt:county_X)
    FILTER (?scenario = nedt:scenario_S)
}

```

CQ9: National demand and contributing counties.

```

SELECT ?scenario ?nationalPeak ?annualEnergy ?annualCO2
        ?county ?countyContribution
WHERE {
    ?nat a nedt:NationalDemandAggregate ;
        nedt:evaluatedUnderScenario ?scenario ;

```

```

        nedt:hasPeakDemandMW ?nationalPeak ;
        nedt:hasAnnualEnergyUse ?annualEnergy ;
        nedt:hasAnnualCO2 ?annualCO2 ;
        nedt:aggregatesDemandFrom ?countyAgg .
?countyAgg a nedt:CountyDemandAggregate ;
        nedt:forCounty ?county ;
        nedt:hasContributionToNationalPeak ?countyContribution .
FILTER (?scenario = nedt:scenario_S)
}
ORDER BY DESC(?countyContribution)

```

CQ10: At-risk stations requiring reinforcement.

```

SELECT ?asset ?scenario ?utilisation ?overloadHours
WHERE {
    ?kpi a nedt:CapacityKPI ;
        nedt:forLVStation ?asset ;
        nedt:evaluatedUnderScenario ?scenario ;
        nedt:hasUtilisation ?utilisation .
    OPTIONAL { ?kpi nedt:hasOverloadHours ?overloadHours . }
    FILTER (?scenario = nedt:scenario_S)
    FILTER (?utilisation >= 0.9)
}
ORDER BY DESC(?utilisation)

```

CQ11: Sensitivity result for a scenario parameter.

```

SELECT ?scenario ?parameter ?marginalEffectPeak ?marginalEffectCO2
WHERE {
    ?sens a nedt:SensitivityResult ;
        nedt:evaluatedUnderScenario ?scenario ;
        nedt:forParameter ?parameter ;
        nedt:hasMarginalEffectOnPeak ?marginalEffectPeak ;
        nedt:hasMarginalEffectOnCO2 ?marginalEffectCO2 .
    FILTER (?scenario = nedt:scenario_S)
}
ORDER BY DESC(?marginalEffectPeak)

```

CQ12: Provenance for a KPI value.

```

SELECT ?kpi ?activity ?dataset ?assumption ?timestamp
WHERE {
    ?kpi a nedt:KPI ;
        prov:wasGeneratedBy ?activity .
    ?activity prov:used ?dataset ;
        prov:generatedAtTime ?timestamp .
    OPTIONAL { ?kpi nedt:hasScenarioAssumption ?assumption . }
    FILTER (?kpi = nedt:kpi_X)
}

```

CQ13: Dwellings assigned to LV transformer T.

```

SELECT ?station ?dwelling ?archetype ?distance
WHERE {
    ?dwelling a nedt:Dwelling ;
        nedt:isServedBy ?station ;
        nedt:instantiatesArchetype ?archetype .
    OPTIONAL { ?dwelling nedt:hasAssignmentDistanceM ?distance . }
    FILTER (?station = nedt:station_T)
}
ORDER BY DESC(?distance)

```

CQ14: MV roll-up from downstream LV stations.

```

SELECT ?parentAsset (SUM(?peakDemand) AS ?rolledUpPeak)
WHERE {
    ?kpi a nedt:CapacityKPI ;
        nedt:forLVStation ?lv ;
        nedt:evaluatedUnderScenario nedt:scenario_S ;
        nedt:hasPeakDemandKW ?peakDemand .
    ?lv nedt:hasParentMVFeeder ?parentAsset .
}
GROUP BY ?parentAsset
ORDER BY DESC(?rolledUpPeak)

```

CQ15: Validation observation linked to a simulated profile.

```

SELECT ?profile ?observation ?measuredValue ?timestamp
WHERE {
    ?profile a nedt:DemandProfile ;
        nedt:validatedAgainstObservation ?observation .
}

```

```

    ?observation a nedt:ValidationObservation ;
                  nedt:hasMeasuredValue ?measuredValue ;
                  nedt:hasTimestamp ?timestamp .
  FILTER (?profile = nedt:profile_X)
}
ORDER BY ?timestamp

```

Appendix F. Cypher Query Skeletons for Neo4j Deployment

The following Cypher skeletons correspond to selected NEDT competency questions and demonstrate that the property graph is queryable through Neo4j without loss of semantic content. Each relationship type maps 1:1 to an RDF property in the T-Box.

CQ1 equivalent — Small areas with the most buildings.

```

MATCH (sa:SmallArea)<-[:HAS_BUILDING]-(b:Building)
RETURN sa.smallAreaRef AS sa_ref,
       count(b) AS n_buildings
ORDER BY n_buildings DESC
LIMIT 5

```

CQ5 equivalent — Small areas with building and LV coverage.

```

MATCH (sa:SmallArea)<-[:HAS_BUILDING]-(b:Building),
      (sa)<-[:SERVES_SMALL_AREA]-(lv:LVStation)
RETURN sa.smallAreaRef AS sa_ref,
       count(DISTINCT b) AS n_buildings,
       count(DISTINCT lv) AS n_lv
ORDER BY n_buildings DESC
LIMIT 5

```

CQ6 equivalent — Buildings served by zero-headroom LV stations.

```

MATCH (c:County)<-[:HAS_COUNTY]-(sa:SmallArea)
      <-[:HAS_BUILDING]-(b:Building)
      -[:NEAREST_LV_STATION]->(lv:LVStation)
WHERE lv.demandAvailableMVA = 0.0
RETURN c.name AS county,
       sa.smallAreaRef AS small_area,
       b.buildingId AS building,
       b.buildTypeFinal AS dwelling_type,

```



```
b.energyRating      AS BER,  
lv.stationName      AS lv_station,  
lv.demandAvailableMVA AS available_capacity_MVA
```